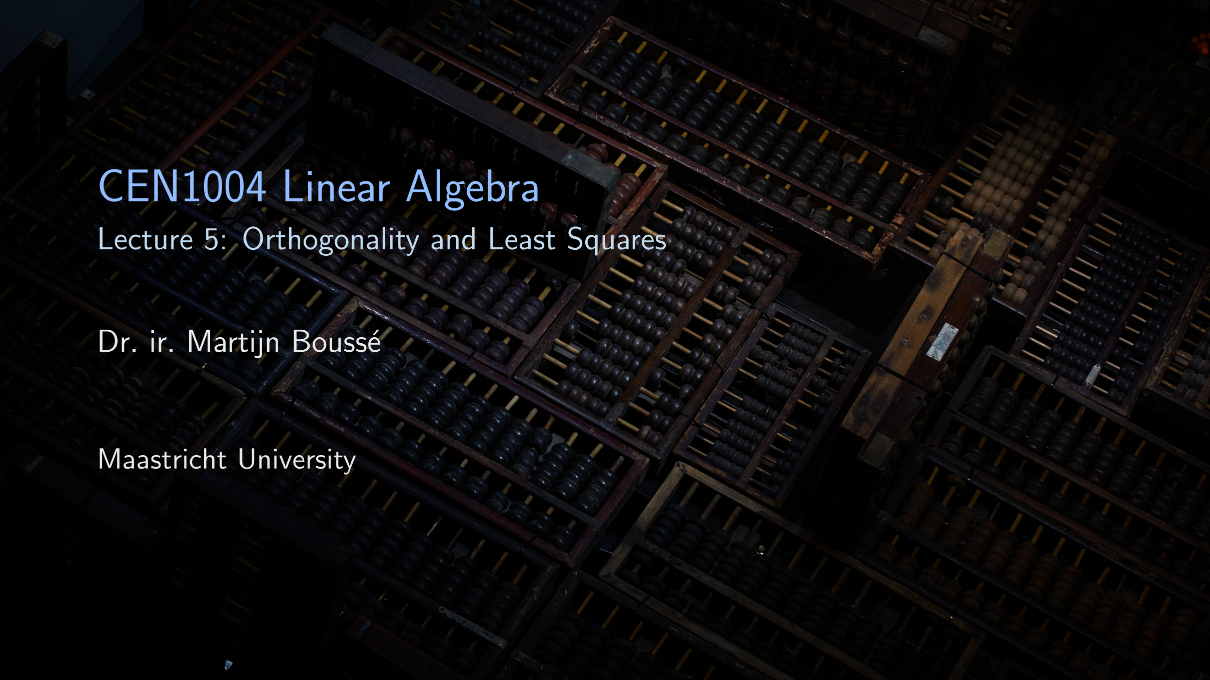

CEN1004 Linear Algebra

Lecture 5: Orthogonality and Least Squares

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Engineers use science, technology, and math to solve problems.

Solving linear systems of equations is fundamental.

Key idea of lecture 1.

Refining your matrix algebra skills is crucial.

Key idea of lecture 2.

Vector spaces provide us with a new, insightful, way to look at linear algebra.

Key idea of lecture 3.

The eigenvalue decomposition is a fundamental tool of linear algebra with various applications.

Finding an approximate solution of a linear system of equations is crucial in many applications.

Key idea of lecture 5. Today! This is, in some sense, an *elaboration* of the key idea of lecture 1.

Recap: Solving linear systems

$$\begin{cases} 1x_1 + 1x_2 & = 0 \\ 0x_2 + 1x_2 & = 1 \end{cases} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{cases} x_1 & = -1 \\ x_2 & = 1 \end{cases}$$

Recap: Solving linear systems

$$\begin{cases} 1x_1 + 1x_2 & = 0 \\ 0x_2 + 1x_2 & = 1 \end{cases} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{cases} x_1 & = -1 \\ x_2 & = 1 \end{cases}$$

The system is consistent, moreover, it has a unique solution!

Recap: Solving linear systems

$$\begin{cases} 1x_1 + 1x_2 & = 0 \\ 0x_2 + 1x_2 & = 1 \\ 2x_1 + 1x_2 & = 0 \end{cases} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Recap: Solving linear systems

$$\begin{cases} 1x_1 + 1x_2 & = 0 \\ 0x_2 + 1x_2 & = 1 \\ 2x_1 + 1x_2 & = 0 \end{cases} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

The system is inconsistent, i.e., there is no solution.

Interpretation: The system is inconsistent because it is overdetermined (or, overconstrained), i.e., we have more equations than variables.

Interpretation: The system is inconsistent because it is overdetermined (or, overconstrained), i.e., we have more equations than variables.

Is there an approximate solution?

Demo: Analyzing an inconsistent system.

See `lecture5_inconsistent_system.m`.

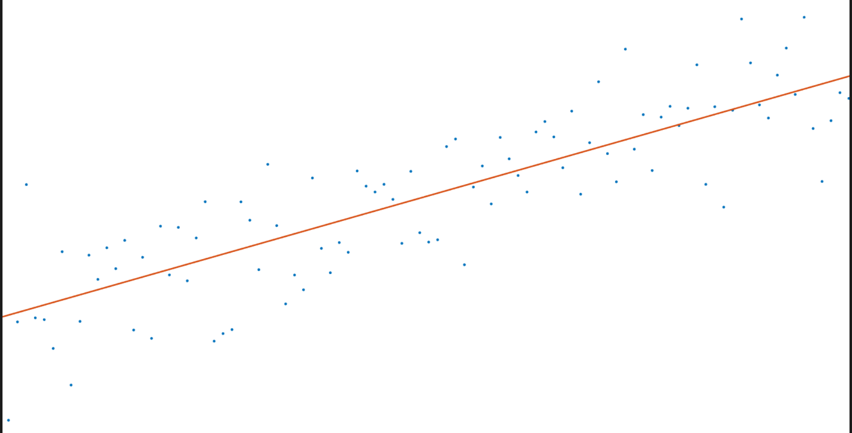
It appears we can find an approximate solution!

In order to define what "approximate" means we need to define two things:

- an expression for distance in $\mathbb{R}^n$,
- and a concept for "nearest" distance (or perpendicularity/orthogonality).

Who cares?

Inconsistent systems are ubiquitous in regression analysis, in which we try to analyze the relationship between a dependent and independent variable.



Regression analysis is crucial for data-driven decision making.

- Comparing ecological footprints — CEN2006 R, R, R, R
- Predicting energy consumption — CEN2008 Energy Systems
- Analyzing impact of food production — CEN2009 Sus. Food Production
- Analyzing smart meters — CEN2018 Circular Business Development
- Predicting plastic production — CEN2020 Biobased Materials

It's very likely you will come in to contact with data-driven decision making in today's data paradise.

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

Orthogonal sets; bases revisited

Orthogonal projection

The Gram–Schmidt process

Least-squares problems

Recap: A (column) vector is a matrix with only one column.

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

Two vectors are equal if and only if their corresponding elements are equal.
For example, when $v_1 = w_1$ and $v_2 = w_2$.

Recap: Two principal vector operations are addition and scalar multiplication.

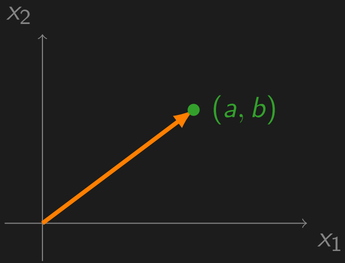
Suppose $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, and $c \in \mathbb{R}$, then

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

and

$$c\mathbf{u} = c \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} cu_1 \\ cu_2 \end{pmatrix}.$$

Recap: Geometrically, a vector $\begin{pmatrix} a \\ b \end{pmatrix}$ corresponds to a point (a, b) in the plane.

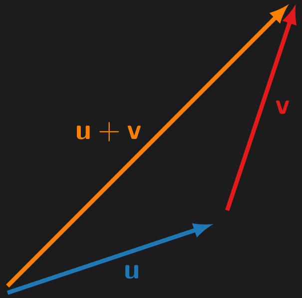


We regard $\mathbb{R}^2$ as the set of all poins in the plane.

Recap: Two principal vector operations are vector addition and scalar multiplication

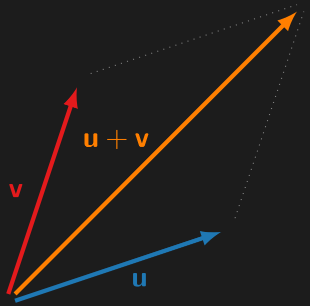
... but now interpreted geometrically!

Recap: Two principal vector operations are vector addition and scalar multiplication



$$\mathbf{u}+\mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

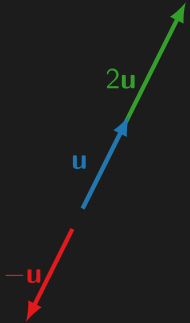
Recap: Two principal vector operations are vector addition and scalar multiplication



$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

The resultant vector is equal to the diagonal of the parallelogram spanned by the constituent vectors.

Recap: Two principal vector operations are vector addition and scalar multiplication



$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}$$

$$c\mathbf{u} = \begin{pmatrix} cu_1 \\ cu_2 \end{pmatrix}$$

Recap: Vector operations have many of the properties of ordinary arithmetic.

Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be vectors and a, b be scalars.

$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$	commutativity
$(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$	associativity
$\mathbf{u}+\mathbf{0}=\mathbf{u}$	identity element
$\mathbf{u}+(-\mathbf{u})=\mathbf{0}$	inverse element
$0\mathbf{u}=\mathbf{0}$	zero element
$1\mathbf{u}=\mathbf{u}$	identity element
$a(b\mathbf{u})=(ab)\mathbf{u}$	associativity
$a(\mathbf{u}+\mathbf{v})=a\mathbf{u}+a\mathbf{v}$	distributivity
$(a+b)\mathbf{u}=a\mathbf{u}+b\mathbf{u}$	distributivity

You can easily verify the properties by using the definitions of vector addition and scalar multiplication.

Recap: A linear combination of vectors involves a weighted sum of the vectors.

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p \in \mathbb{R}^n$ and scalars $c_1, c_2, \cdots, c_p \in R$, the vector $\mathbf{y}$ defined as

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p$ with weights $c_1, c_2, \cdots, c_p$.

The inner product of two vectors $\mathbf{u}$ and $\mathbf{v}$ is defined as $\mathbf{u}^\mathsf{T}\mathbf{v}$.

$$\mathbf{u}^\mathsf{T}\mathbf{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

The inner product is also known as the dot product, which is typically denoted by $\mathbf{u} \bullet \mathbf{v}$.

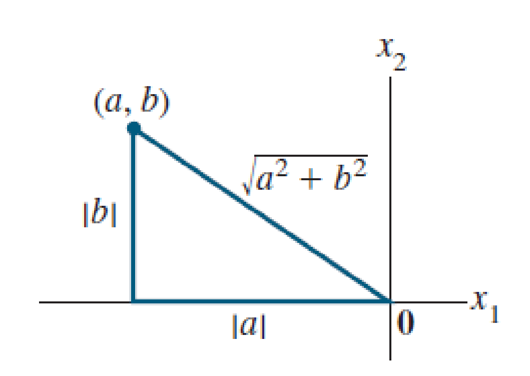
The properties of the inner product follow immediately from the properties of the transpose, see lecture 2.

Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $c \in \mathbb{R}$, then:

- $\mathbf{u} \bullet \mathbf{v} = \mathbf{v} \bullet \mathbf{u}$ — commutativity
- $(\mathbf{u} + \mathbf{v}) \bullet \mathbf{w} = \mathbf{u} \bullet \mathbf{w} + \mathbf{v} \bullet \mathbf{w}$ — distributivity
- $(c\mathbf{u}) \bullet \mathbf{v} = c(\mathbf{u} \bullet \mathbf{v}) = \mathbf{u} \bullet (c\mathbf{v})$ — associativity
- $\mathbf{u} \bullet \mathbf{u} \geq 0$, and $\mathbf{u} \bullet \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Useful rule: $(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots c_n\mathbf{u}_n) \bullet \mathbf{w} = c_1(\mathbf{u}_1 \bullet \mathbf{w}) + c_2(\mathbf{u}_2 \bullet \mathbf{w}) + \cdots + c_n(\mathbf{u}_n \bullet \mathbf{w})$.

What is the length of a vector?



This idea extends to higher dimensions as well, see CEN1008 Multivariable Calculus.

The length (or norm) of a vector $\mathbf{v}$ is defined as the square root of the inner product of $\mathbf{v}$ with itself.

$$||\mathbf{v}|| = \sqrt{\mathbf{v} \bullet \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad \text{and} \quad ||\mathbf{v}||^2 = \mathbf{v} \bullet \mathbf{v}$$

Note that $||c\mathbf{v}|| = |c|||\mathbf{v}||$.

A unit vector is a vector with length 1.

Consider the vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \mathbf{v} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

with norms

$$||\mathbf{u}|| = \sqrt{2} \qquad ||\mathbf{v}|| = 1$$

Hence, $\mathbf{u}$ is not a unit vector and $\mathbf{v}$ is a unit vector.

We can obtain a unit vector (i.e., a vector of length 1) by normalizing the vector.

Normalizing a vector $\mathbf{v}$ means dividing the vector by its length $||\mathbf{v}||$, i.e., $\frac{\mathbf{v}}{||\mathbf{v}||}$.

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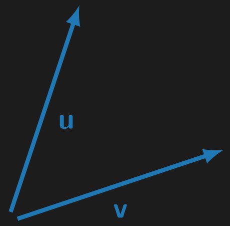
with norms

$$||\mathbf{u}|| = \sqrt{2} \qquad ||\mathbf{v}|| = 1$$

We say that both vectors are *in the same direction*.

The distance between $\mathbf{u}$ and $\mathbf{v}$, written as $\text{dist}(\mathbf{u}, \mathbf{v})$, is the length of the vector $\mathbf{u} - \mathbf{v}$, i.e., $\text{dist}(\mathbf{u}, \mathbf{v}) = ||\mathbf{u} - \mathbf{v}||$.



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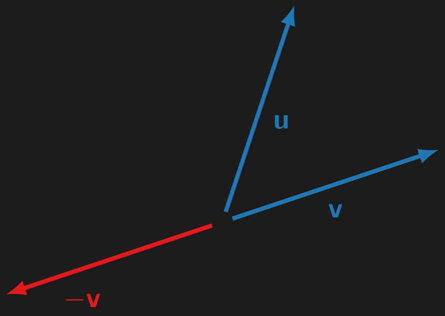

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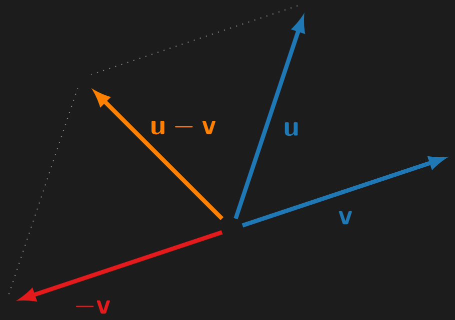

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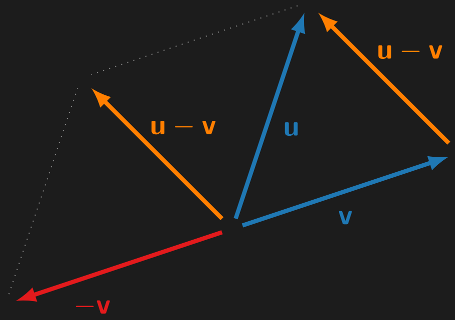


Example: Compute the distance between $\mathbf{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Compute the difference between $\mathbf{u}$ and $\mathbf{v}$:

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

Compute the length of the resulting vector:

$$||\mathbf{u} - \mathbf{v}|| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$

Hence, $\text{dist}(\mathbf{u}, \mathbf{v}) = \sqrt{17}$.

It appears we can find an approximate solution!

In order to define what "approximate" means we need to define two things:

- an expression for distance in $\mathbb{R}^n$,
- and a concept for "nearest" distance (or perpendicularity/orthogonality).

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

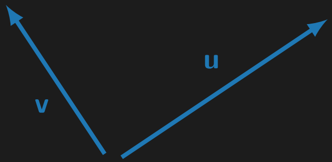
Orthogonal sets; bases revisited

Orthogonal projection

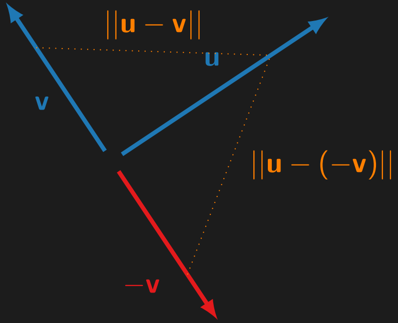
The Gram–Schmidt process

Least-squares problems

Example: Perpendicular vectors



Example: Perpendicular vectors



Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

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Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

$$\begin{aligned} \operatorname{dist}(\mathbf{u}, -\mathbf{v}) &= ||\mathbf{u} - (-\mathbf{v})|| = ||\mathbf{u} + \mathbf{v}|| \\ &= \sqrt{(\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v})} \end{aligned}$$

Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

$$\begin{aligned}[\operatorname{dist}(\mathbf{u}, -\mathbf{v})]^2 &= ||\mathbf{u} - (-\mathbf{v})||^2 = ||\mathbf{u} + \mathbf{v}||^2 \\ &= (\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v})\end{aligned}$$

Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

$$\begin{aligned}
[\text{dist}(\mathbf{u}, -\mathbf{v})]^2 = ||\mathbf{u} - (-\mathbf{v})||^2 = ||\mathbf{u} + \mathbf{v}||^2 \\
= (\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v}) \\
= \mathbf{u} \bullet (\mathbf{u} + \mathbf{v}) + \mathbf{v} \bullet (\mathbf{u} + \mathbf{v})
\end{aligned}$$

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&= \mathbf{u} \bullet \mathbf{u} + \mathbf{u} \bullet \mathbf{v} + \mathbf{v} \bullet \mathbf{u} + \mathbf{v} \bullet \mathbf{v}
\end{aligned}$$

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&= ||\mathbf{u}||^2 + ||\mathbf{v}||^2 + 2\mathbf{u} \bullet \mathbf{v}
\end{aligned}$$

Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

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$$[\mathrm{dist}(\mathbf{u}, \mathbf{v})] = ||\mathbf{u}||^2 + ||\mathbf{v}||^2 - 2\mathbf{u} \bullet \mathbf{v}$$

Two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^n$ are orthogonal (to each other) if $\mathbf{u} \bullet \mathbf{v} = 0$.

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\end{aligned}$$

$$[\mathrm{dist}(\mathbf{u}, \mathbf{v})] = ||\mathbf{u}||^2 + ||\mathbf{v}||^2 - 2\mathbf{u} \bullet \mathbf{v}$$

The distances are equal if and only if $2\mathbf{u} \bullet \mathbf{v} = -2\mathbf{u} \bullet \mathbf{v}$, which happens if and only if $\mathbf{u} \bullet \mathbf{v} = 0$.

Now you try!

The standard unit vectors in $\mathbb{R}^2$ are orthogonal. T/F?

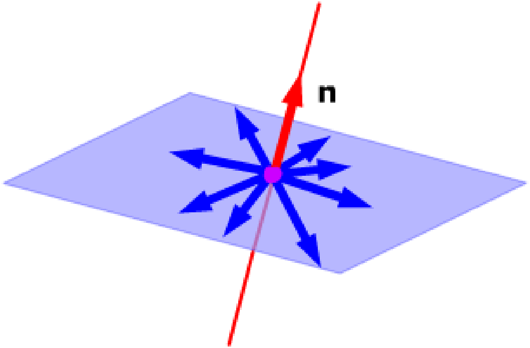
Now you try!

The standard unit vectors in $\mathbb{R}^2$ are orthogonal. T/F?

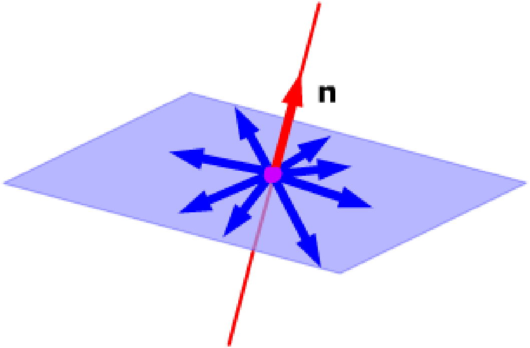
True! The inner product is equal to zero:

$$\mathbf{e}_1 \bullet \mathbf{e}_2 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

If a vector $\mathbf{n}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then we say that $\mathbf{n}$ is orthogonal to W.



The set of all vectors $\mathbf{n}$ that are orthogonal to W is called the orthogonal complement of W, denoted by $W^\perp$ (and pronounced "W perpendicular").



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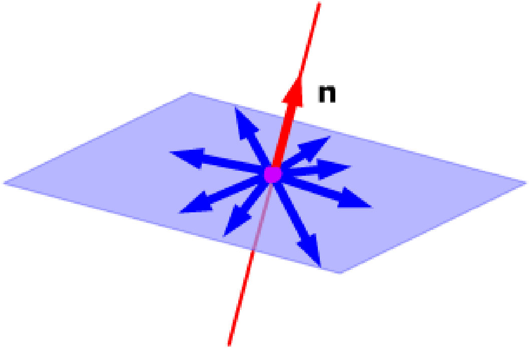

In the example above, W is two-dimensional (represented by a plane) and $W^{\perp}$ is one-dimensional (represented by a line L). note that $L = W^{\perp}$ and $W = L^{\perp}$.

We need two more facts:

- A vector $\mathbf{x}$ is in $W^{\perp}$ if and only if $\mathbf{x}$ is orthogonal to every vector in a set that spans W.
- $W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Let's revisit familiar subspaces with this new terminology.

Recap: The column space of an $m \times n$ matrix $\mathbf{A}$, written as Col $\mathbf{A}$, is the set of all linear combinations of the columns of $\mathbf{A}$.

$$\text{If } \mathbf{A} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}, \text{ then Col } \mathbf{A} = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

See lecture 3.

Recap: The row space is the set of all linear combinations of the rows of a matrix.

$$\text{Row } \mathbf{A} = \text{Col } \mathbf{A}^{\mathsf{T}}$$

See lecture 3.

Recap: The null space of an $m \times n$ matrix $\mathbf{A}$, written as Nul $\mathbf{A}$, is the set of all solutions of the homogeneous equation $\mathbf{Ax} = \mathbf{0}$.

In set notation, Nul $\mathbf{A} = \{\mathbf{x} \ : \ \mathbf{x}$ is in $\mathbb{R}^n$ and $\mathbf{Ax} = \mathbf{0}\}$.

See lecture 3.

Suppose $\mathbf{x} \in \text{Nul } \mathbf{A}$, then we know that $\mathbf{A}\mathbf{x} = \mathbf{0}$, which means that

$$\mathbf{a}_k^\top \mathbf{x} = 0,$$

i.e., $\mathbf{x}$ is orthogonal to each row of $\mathbf{A}$ (which span Row $\mathbf{A}$).

Suppose $\mathbf{x} \in \text{Nul } \mathbf{A}$, then we know that $\mathbf{A}\mathbf{x} = \mathbf{0}$, which means that

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i.e., $\mathbf{x}$ is orthogonal to each row of $\mathbf{A}$ (which span Row $\mathbf{A}$).

Suppose that $\mathbf{x}$ is orthogonal to Row $\mathbf{A}$, then we have that

$$\mathbf{a}_k^\top \mathbf{x} = 0,$$

i.e., $\mathbf{A}\mathbf{x} = 0$, hence, $\mathbf{x} \in \text{Nul } \mathbf{A}$.

Suppose $\mathbf{x} \in \text{Nul } \mathbf{A}$, then we know that $\mathbf{Ax} = \mathbf{0}$, which means that

$$\mathbf{a}_k^\mathsf{T}\mathbf{x} = 0,$$

i.e., $\mathbf{x}$ is orthogonal to each row of $\mathbf{A}$ (which span Row $\mathbf{A}$).

Suppose that $\mathbf{x}$ is orthogonal to Row $\mathbf{A}$, then we have that

$$\mathbf{a}_k^\mathsf{T}\mathbf{x} = 0,$$

i.e., $\mathbf{Ax} = 0$, hence, $\mathbf{x} \in \text{Nul } \mathbf{A}$.

This means that $(\text{Row } \mathbf{A})^\perp = \text{Nul } \mathbf{A}$.

We know that $(\text{Row } \mathbf{A})^\perp = \text{Nul } \mathbf{A}$ holds true for any matrix, so it also holds for $\mathbf{A}^\mathsf{T}$.

In other words,

$$\begin{aligned}
(\text{Row } \mathbf{A})^\perp &= \text{Nul } \mathbf{A} \\
(\text{Row } \mathbf{A}^\mathsf{T})^\perp &= \text{Nul } \mathbf{A}^\mathsf{T} \\
(\text{Col } \mathbf{A})^\perp &= \text{Nul } \mathbf{A}^\mathsf{T}
\end{aligned}$$

The fundamental subspaces of a matrix $\mathbf{A}$ are related to each other in terms of orthogonal complements.

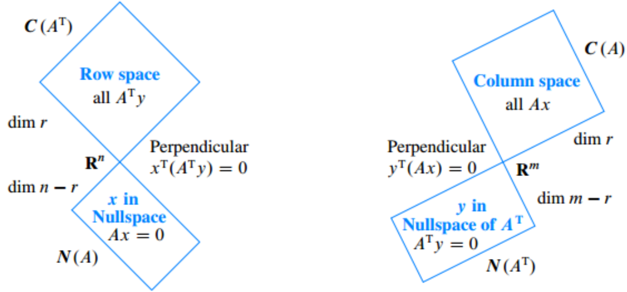


Figure 1: Dimensions and orthogonality for any m by n matrix A of rank r.

This is an important figure for the exam! You can make some funky T/F statements about this.

Learning objectives

1. Solve linear systems of equations.
2. Compute matrix factorizations.
3. Solve and compute various linear algebra problems using Matlab.
4. Communicate and reason about linear algebra problems algebraically and geometrically by interleaving various definitions, theorems, and properties about vectors and matrices, linear systems and factorizations, eigenvalues and eigenvectors, singular values and singular vectors, linear transformations, orthogonality, determinants, etc.
5. Understand that many, seemingly disconnected, problems from various disciplines reduce to linear algebra problems.
6. Perform calculations with complex numbers.

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

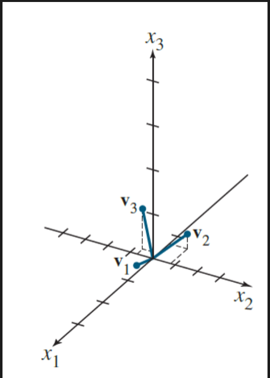
Orthogonal sets; bases revisited

Orthogonal projection

The Gram–Schmidt process

Least-squares problems

A set of vectors is orthogonal if each pair of distinct vectors from the set is orthogonal.



The set $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\} \in \mathbb{R}^n$ is orthogonal if $\mathbf{u}_i \bullet \mathbf{u}_j = 0$, whenever $i \neq j$.

A set of vectors is orthonormal if the set is an orthogonal set of unit vectors.

Let's revisit bases with this new terminology.

Recap: A basis represents a subspace without any unnecessary vectors.

Let H be a subspace of a vector space V. A set of vectors $\mathcal{B}$ in V is a basis for H if

1. $\mathcal{B}$ is a linearly independent set, and
2. the subspace spanned by $\mathcal{B}$ coincides with H, i.e.,

$$H = \text{Span } \mathcal{B}$$

See lecture 3.

An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Let H be a subspace of a vector space V. A set of vectors $\mathcal{B}$ in V is an orthogonal basis for H if

1. $\mathcal{B}$ is a linearly independent set, and
2. $\mathcal{B}$ is an orthogonal set, and
3. the subspace spanned by $\mathcal{B}$ coincides with H, i.e.,

$$H = \text{Span } \mathcal{B}$$

An orthogonal basis is much nicer than other bases!

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y} \in W$, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \bullet \mathbf{u}_j}{\mathbf{u}_j \bullet \mathbf{u}_j} \quad 1 \leq j \leq p$$

$$
\begin{aligned}
\mathbf{y} \bullet \mathbf{u}_1 &= (c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p) \bullet \mathbf{u}_1 \\
&= c_1\mathbf{u}_1 \bullet \mathbf{u}_1 + c_2\mathbf{u}_2 \bullet \mathbf{u}_1 + \cdots + c_p\mathbf{u}_p \bullet \mathbf{u}_1 \\
&= c_1\mathbf{u}_1 \bullet \mathbf{u}_1
\end{aligned}
$$

The last equality holds because the vectors are orthogonal so all inner products are zero except the first.

Example: Express a vector in an orthogonal basis.

$$\text{Express } \mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \text{ in } S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}.$$

Example: Express a vector in an orthogonal basis.

$$\text{Express } \mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \text{ in } S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}.$$

Let's first double-check that the set is orthogonal:

Example: Express a vector in an orthogonal basis.

Express $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ in $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}$.

Let's first double-check that the set is orthogonal:

$$\begin{aligned} \mathbf{u}_1 \bullet \mathbf{u}_2 &= 3(-1) + 1(2) + 1(1) = 0 \\ \mathbf{u}_1 \bullet \mathbf{u}_3 &= 3(-1/2) + 1(-2) + 1(7/2) = 0 \\ \mathbf{u}_2 \bullet \mathbf{u}_3 &= -1(-1/2) + 2(-2) + 1(7/2) = 0 \end{aligned}$$

Example: Express a vector in an orthogonal basis.

$$\text{Express } \mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \text{ in } S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}.$$

According to the previous theorem, we have:

$$\mathbf{y} = \frac{\mathbf{y} \bullet \mathbf{u}_1}{\mathbf{u}_1 \bullet \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y} \bullet \mathbf{u}_2}{\mathbf{u}_2 \bullet \mathbf{u}_2}\mathbf{u}_2 + \frac{\mathbf{y} \bullet \mathbf{u}_3}{\mathbf{u}_3 \bullet \mathbf{u}_3}\mathbf{u}_3$$

Example: Express a vector in an orthogonal basis.

Express $\mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ in $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}$.

According to the previous theorem, we have:

$$\begin{aligned}
\mathbf{y} &= \frac{\mathbf{y} \bullet \mathbf{u}_1}{\mathbf{u}_1 \bullet \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{y} \bullet \mathbf{u}_2}{\mathbf{u}_2 \bullet \mathbf{u}_2}\mathbf{u}_2 + \frac{\mathbf{y} \bullet \mathbf{u}_3}{\mathbf{u}_3 \bullet \mathbf{u}_3}\mathbf{u}_3 \\
&= \frac{11}{11}\mathbf{u}_1 + \frac{-12}{6}\mathbf{u}_2 + \frac{-33}{33/2}\mathbf{u}_3 \\
&= \mathbf{u}_1 - 2\mathbf{u}_2 - 2\mathbf{u}_3
\end{aligned}$$

Example: Express a vector in an orthogonal basis.

$$\text{Express } \mathbf{y} = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix} \text{ in } S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix} \right\}.$$

Recall our method from lecture 1: Row reduce the augmented matrix

$$\begin{bmatrix} 3 & -1 & -1/2 & 6 \\ 1 & 2 & -2 & 1 \\ 1 & 1 & 7/2 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

An orthogonal basis is much nicer than other bases!

Recap: An orthogonal basis is much nicer than other bases!

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y} \in W$, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y} \bullet \mathbf{u}_j}{\mathbf{u}_j \bullet \mathbf{u}_j} \quad 1 \leq j \leq p$$

$\Rightarrow$ We want to interpret this theorem geometrically!

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

Orthogonal sets; bases revisited

Orthogonal projection

The Gram–Schmidt process

Least-squares problems

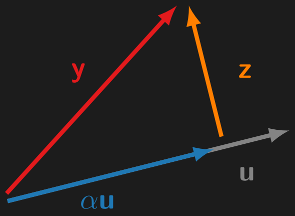
Decompose $\mathbf{y} \in \mathbb{R}^n$ into a sum of two vectors

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

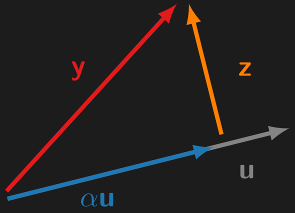
where

- $\hat{\mathbf{y}} = \alpha \mathbf{u}$
- $\mathbf{z} \perp \mathbf{u}$

with $\mathbf{u} \in \mathbb{R}^n$ a nonzero vector.

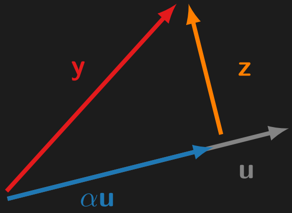


Let's use what we know!



Let's use what we know!

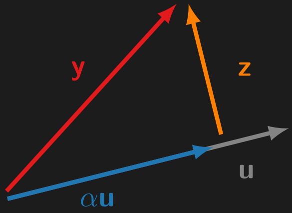
The vector $\mathbf{z}$ is orthogonal to $\mathbf{u}$ if and only if



Let's use what we know!

The vector **z** is orthogonal to **u** if and only if

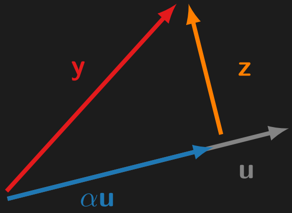
$$\mathbf{z} \bullet \mathbf{u} = 0$$



Let's use what we know!

The vector $\mathbf{z}$ is orthogonal to $\mathbf{u}$ if and only if

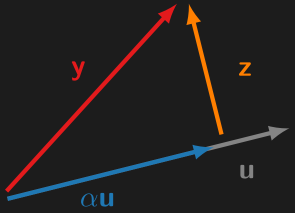
$$\mathbf{z} \bullet \mathbf{u} = 0$$
$$(\mathbf{y} - \alpha\mathbf{u}) \bullet \mathbf{u} = 0$$



Let's use what we know!

The vector $\mathbf{z}$ is orthogonal to $\mathbf{u}$ if and only if

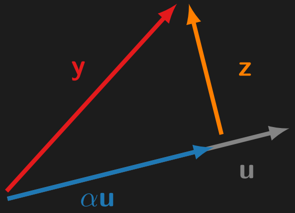
$$\begin{aligned} \mathbf{z} \bullet \mathbf{u} &= 0 \\ (\mathbf{y} - \alpha\mathbf{u}) \bullet \mathbf{u} &= 0 \\ \mathbf{y} \bullet \mathbf{u} - \alpha\mathbf{u} \bullet \mathbf{u} &= 0 \end{aligned}$$



Let's use what we know!

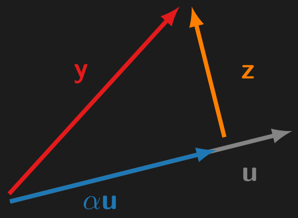
The vector $\mathbf{z}$ is orthogonal to $\mathbf{u}$ if and only if

$$\begin{aligned}
\mathbf{z} \bullet \mathbf{u} &= 0 \\
(\mathbf{y} - \alpha\mathbf{u}) \bullet \mathbf{u} &= 0 \\
\mathbf{y} \bullet \mathbf{u} - \alpha\mathbf{u} \bullet \mathbf{u} &= 0 \\
\alpha &= \frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}
\end{aligned}$$



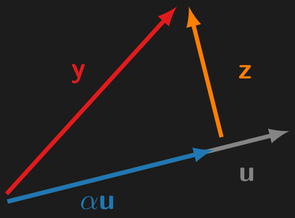
We can decompose $\mathbf{y}$ along a vector $\mathbf{u}$ and orthogonal to a vector $\mathbf{u}$ as follows:

$$\begin{aligned}\mathbf{y} &= \alpha\mathbf{u} + \mathbf{z} \\ &= \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u} + \mathbf{z}\end{aligned}$$



We can decompose $\mathbf{y}$ along a vector $\mathbf{u}$ and orthogonal to a vector $\mathbf{u}$ as follows:

$$\begin{aligned}\mathbf{y} &= \alpha\mathbf{u} + \mathbf{z} \\ &= \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u} + \mathbf{z}\end{aligned}$$



We call $\alpha\mathbf{u}$ the orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$, written as,

$$\text{proj}_{\mathbf{u}}\mathbf{y} = \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u}$$

Example: Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

Compute the orthogonal projection $\hat{\mathbf{y}}$:

$$\begin{aligned} \hat{\mathbf{y}} &= \mathrm{proj}_{\mathbf{u}}\mathbf{y} \\ &= \left(\frac{\mathbf{y} \bullet \mathbf{u}}{\mathbf{u} \bullet \mathbf{u}}\right)\mathbf{u} \\ &= \left(\frac{\begin{bmatrix} 7 & 6 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}}{\begin{bmatrix} 4 & 2 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}}\right)\begin{bmatrix} 4 \\ 2 \end{bmatrix} \\ &= \left(\frac{40}{20}\right)\begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} \end{aligned}$$

Example: Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

The component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is given by

$$\begin{aligned} \mathbf{z} &= \mathbf{y} - \hat{\mathbf{y}} \\ &= \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ 2 \end{bmatrix} \end{aligned}$$

Example: Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

The component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is given by

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Sanity check:

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}.$$

Example: Find the orthogonal projection of $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ onto $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$.

The component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is given by

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Sanity check:

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}.$$

Yet another sanity check:

$$\hat{\mathbf{y}} \bullet \mathbf{z} = \begin{bmatrix} 8 & 4 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 0$$

The formula for orthogonal projection is exactly the same as the one we found for the coefficients in an orthogonal basis.

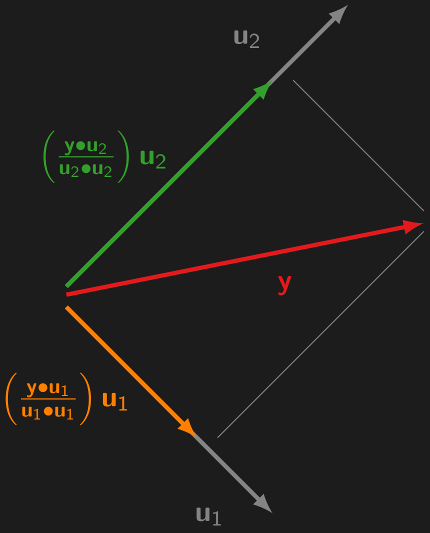
Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y} \in W$, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

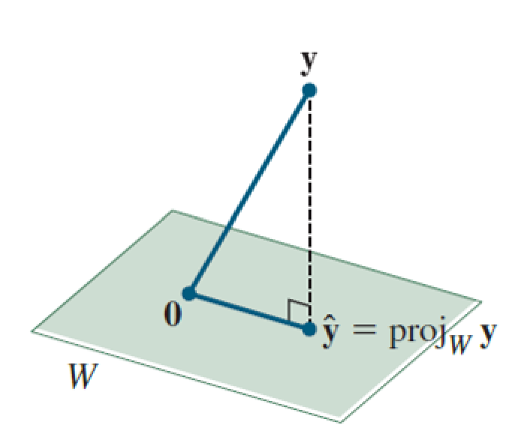
$$c_j = \frac{\mathbf{y} \bullet \mathbf{u}_j}{\mathbf{u}_j \bullet \mathbf{u}_j} \quad 1 \leq j \leq p$$

Geometrically, expressing a vector in an orthogonal basis is the same as projecting the vector orthogonally onto the basis vectors.



This is a crucial insight!

Instead of projecting a vector onto a vector, how can we project a vector onto a plane?



The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

In fact, if $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \left(\frac{\mathbf{y} \bullet \mathbf{u}_1}{\mathbf{u}_1 \bullet \mathbf{u}_1}\right)\mathbf{u}_1 + \cdots + \left(\frac{\mathbf{y} \bullet \mathbf{u}_p}{\mathbf{u}_p \bullet \mathbf{u}_p}\right)\mathbf{u}_p$$

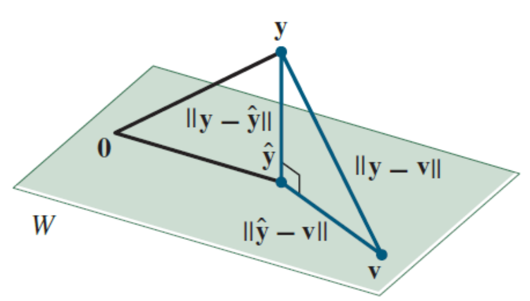
and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

The proof is similar to the one for a line, see Section 6.3.

Demo: Orthogonal projection onto a plane.

Click here.

Best Approximation Theorem: The "best" approximation to $\mathbf{y}$ by elements of W is equal to the orthogonal projection of $\mathbf{y}$ onto W.



For all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$, we have $||\mathbf{y} - \hat{\mathbf{y}}|| < ||\mathbf{y} - \mathbf{v}||$.

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6. Perform calculations with complex numbers.

Up till now, we were always given an orthogonal set, but how can we construct one?

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

Orthogonal sets; bases revisited

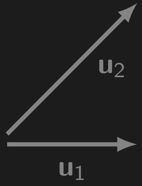
Orthogonal projection

The Gram–Schmidt process

Least-squares problems

Example: Construct an orthogonal basis given a set of vectors in $\mathbb{R}^2$.

Let $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ be a basis for $\mathbb{R}^2$.



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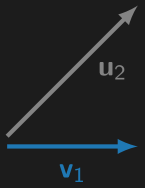
Choose $\mathbf{v}_1 = \mathbf{u}_1$.

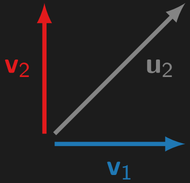


Example: Construct an orthogonal basis given a set of vectors in $\mathbb{R}^2$.

Let $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ be a basis for $\mathbb{R}^2$.

Choose $\mathbf{v}_1 = \mathbf{u}_1$.

Construct $\mathbf{v}_2$ by subtracting the part of $\mathbf{u}_2$ *along* $\mathbf{v}_1$ from $\mathbf{u}_2$,



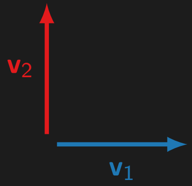
Example: Construct an orthogonal basis given a set of vectors in $\mathbb{R}^2$.

Let $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ be a basis for $\mathbb{R}^2$.

Choose $\mathbf{v}_1 = \mathbf{u}_1$.

Construct $\mathbf{v}_2$ by subtracting the part of $\mathbf{u}_2$ *along* $\mathbf{v}_1$ from $\mathbf{u}_2$, i.e.,

$$\begin{aligned} \mathbf{v}_2 &= \mathbf{u}_2 - \text{proj}_{\mathbf{v}_1} \mathbf{u}_2 \\ &= \mathbf{u}_2 - \left(\frac{\mathbf{u}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1} \right) \mathbf{v}_1 \end{aligned}$$



Demo: Orthogonalizing a set in 3D.

Click here.

The Gram–Schmidt process is a simple algorithm for producing an orthogonal basis for any nonzero subspace of $\mathbb{R}^n$.

Given a basis $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1, \\
\mathbf{v}_2 &= \mathbf{x}_2 - \left(\frac{\mathbf{x}_2 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\right)\mathbf{v}_1, \\
\mathbf{v}_3 &= \mathbf{x}_3 - \left(\frac{\mathbf{x}_3 \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\right)\mathbf{v}_1 - \left(\frac{\mathbf{x}_3 \bullet \mathbf{v}_2}{\mathbf{v}_2 \bullet \mathbf{v}_2}\right)\mathbf{v}_2, \\
&\vdots \\
\mathbf{v}_p &= \mathbf{x}_p - \left(\frac{\mathbf{x}_p \bullet \mathbf{v}_1}{\mathbf{v}_1 \bullet \mathbf{v}_1}\right)\mathbf{v}_1 - \cdots - \left(\frac{\mathbf{x}_p \bullet \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \bullet \mathbf{v}_{p-1}}\right)\mathbf{v}_{p-1}.
\end{aligned}$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is an orthogonal basis for W. In addition

$$\operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \operatorname{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_k\} \quad \text{for} \quad 1 \leq k \leq p.$$

The Gram–Schmidt process (with normalizations) amounts to a new factorization.

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$$\mathbf{a}_k = r_{1k}\mathbf{u}_1 + \cdots + r_{kk}\mathbf{u}_k + 0\mathbf{u}_{k+1} + \cdots + 0\mathbf{u}_n.$$

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Stack the orthonormal basis in a matrix $\mathbf{Q} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix}$ and put the r-coefficients in vectors $\mathbf{r}_k^T = \begin{bmatrix} r_{1k} & \cdots & r_{kk} & 0 & \cdots & 0 \end{bmatrix}$ for $1 \leq k \leq n$.

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Then we can summarize everything in a QR factorization of $\mathbf{A}$, i.e., $\mathbf{A} = \mathbf{QR}$.

If $\mathbf{A}$ is an $m \times n$ matrix with linearly independent columns, then $\mathbf{A}$ can be factored as

$$\mathbf{A} = \mathbf{QR},$$

where

- $\mathbf{Q}$ is an $m \times n$ matrix whose columns form an orthonormal basis for Col $\mathbf{A}$, and,
- $\mathbf{R}$ is an upper triangular invertible matrix with positive diagonal entries.

The QR factorization is a fundamental matrix decomposition that is widely used in numerical algorithms such as algorithms to solve linear equations, see further.

Procedure: Compute a QR factorization of a matrix.

Given an $m \times n$ matrix $\mathbf{A}$.

1. Orthonormalize the columns to obtain $\mathbf{Q}$ (by Gram–Schmidt).
2. Compute $\mathbf{R} = \mathbf{Q}^T\mathbf{A}$.

Note that $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}$ because the columns of $\mathbf{Q}$ are orthonormal. Hence, left-multiplying both sides of the QR factorization by $\mathbf{Q}^\mathsf{T}$ leads to $\mathbf{Q}^\mathsf{T}\mathbf{A} = \mathbf{Q}^T(\mathbf{QR}) = \mathbf{IR} = \mathbf{R}$.

Learning objectives

1. Solve linear systems of equations.
2. Compute matrix factorizations such as the LU factorization, QR factorization, eigenvalue decomposition, and singular value decomposition.
3. Solve and compute various linear algebra problems using Matlab.
4. Communicate and reason about linear algebra problems algebraically and geometrically.
5. Understand that many, seemingly disconnected, problems from various disciplines reduce to linear algebra problems.
6. Perform calculations with complex numbers.

Distance in $\mathbb{R}^n$; vectors revisited

Orthogonality; subspaces revisited

Orthogonal sets; bases revisited

Orthogonal projection

The Gram–Schmidt process

Least-squares problems

Recap: Is there an approximate solution for inconsistent systems?

$$\begin{cases} 1x_1 + 1x_2 & = 0 \\ 0x_2 + 1x_2 & = 1 \\ 2x_1 + 1x_2 & = 0 \end{cases} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

When no solution exists, the best we can do is to find an $\hat{\mathbf{x}}$ that makes $\mathbf{A}\hat{\mathbf{x}}$ as close as possible to $\mathbf{b}$.

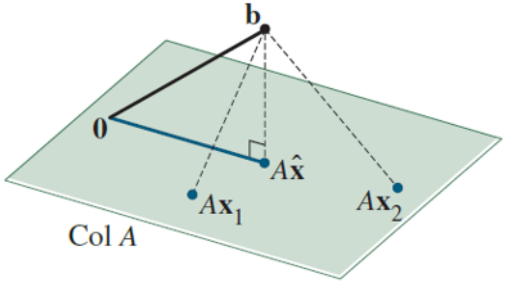


FIGURE 1 The vector $\mathbf{b}$ is closer to $A\hat{\mathbf{x}}$ than to $A\mathbf{x}$ for other $\mathbf{x}$.

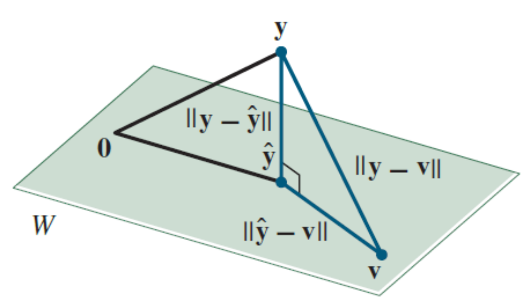
If $\mathbf{A}$ is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a least-squares solution of $\mathbf{Ax} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$||\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}|| \leq ||\mathbf{b} - \mathbf{Ax}||$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

The term "least-squares" arises from the fact that $||\mathbf{b} - \mathbf{Ax}||$ is the square root of a sum of squares, see definition of norm/length.

Recap: Best Approximation Theorem: The "best" approximation to $\mathbf{y}$ by elements of W is equal to the orthogonal projection of $\mathbf{y}$ onto W.



For all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$, we have $||\mathbf{y} - \hat{\mathbf{y}}|| < ||\mathbf{y} - \mathbf{v}||$.

The best approximation theorem tells us that the closest point $\hat{\mathbf{b}}$ to $\mathbf{b}$ is given by:

$$\hat{\mathbf{b}} = \mathrm{proj}_{\mathrm{Col}\ \mathbf{A}}\mathbf{b}.$$

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By definition, $\hat{\mathbf{b}}$ is in Col $\mathbf{A}$, hence, there exists a $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\mathbf{A}\hat{\mathbf{x}} = \hat{\mathbf{b}}.$$

In other words, the system should be consistent because $\hat{\mathbf{b}}$ is in Col $\mathbf{A}$ (by IMT).

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Note that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to Col $\mathbf{A}$.

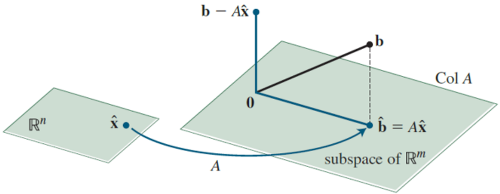

FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

The fact that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to Col $\mathbf{A}$ can be expressed as:

$$\mathbf{a}_j \bullet (\mathbf{b} - \hat{\mathbf{b}}) = 0 \quad \text{for} \quad 1 \leq j \leq n$$

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\mathbf{a}_j \bullet (\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) &= 0 \quad \text{for} \quad 1 \leq j \leq n \\
\mathbf{A}^{\mathsf{T}}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) &= 0 \\
\mathbf{A}^{\mathsf{T}}\mathbf{b} - \mathbf{A}^{\mathsf{T}}\mathbf{A}\hat{\mathbf{x}} &= 0
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\mathbf{A}^T(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) &= 0 \\
\mathbf{A}^T\mathbf{b} - \mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} &= 0 \\
\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} &= \mathbf{A}^T\mathbf{b} \quad \text{aka the normal equations}
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The least-squares solution of $\mathbf{Ax} = \mathbf{b}$ is equal to the solution of the normal equations.

Example: Find the least-squares solution of an inconsistent system.

$$\begin{cases} 1x_1 + 1x_2 &= 0 \\ 0x_2 + 1x_2 &= 1 \\ 2x_1 + 1x_2 &= 0 \end{cases} \rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 2 & 1 \end{bmatrix}}_{=\mathbf{A}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{=\mathbf{b}}$$

The normal equations are given by

$$\mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 2 & 1 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \begin{cases} x_1 &= -3 \\ x_2 &= 5 \end{cases}$$

Let $\mathbf{A}$ be an $m \times n$ matrix. The following statements are logically equivalent:

(a) The equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.

(b) The columns of $\mathbf{A}$ are linearly independent.

(c) The matrix $\mathbf{A}^\top\mathbf{A}$ is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$\hat{\mathbf{x}} = \left(\mathbf{A}^\top\mathbf{A}\right)^{-1}\mathbf{A}^\top\mathbf{b}.$$

Procedure: Least-squares solution via normal equations.

Given an $m \times n$ matrix $\mathbf{A}$.

1. Solve $\mathbf{A}^\top \mathbf{A}\mathbf{x} = \mathbf{A}^\top \mathbf{b}$

Procedure: Least-squares solution via QR factorization.

Given an $m \times n$ matrix $\mathbf{A}$.

1. Compute a QR decomposition of $\mathbf{A} = \mathbf{QR}$.
2. Solve $\mathbf{Rx} = \mathbf{Q}^{\mathsf{T}}\mathbf{b}$.

$$
\begin{aligned}
\mathbf{Ax} &= \mathbf{b} \\
\mathbf{QRx} &= \mathbf{b} \\
\mathbf{Q}^{\mathsf{T}}\mathbf{QRx} &= \mathbf{Q}^{\mathsf{T}}\mathbf{b} \\
\mathbf{Rx} &= \mathbf{Q}^{T}\mathbf{b}
\end{aligned}
$$

Click here.

Finding an approximate solution of a linear system of equations is crucial in many applications.

Learning objectives

1. Solve linear systems of equations using various methods such as Gaussian elimination, LU factorization, the least-squares method, QR factorization, (pseudo)-inverse matrix, and Cramer's rule.
2. Compute matrix factorizations such as the LU factorization, QR factorization, eigenvalue decomposition, and singular value decomposition.
3. Solve and compute various linear algebra problems using Matlab.
4. Communicate and reason about linear algebra problems algebraically and geometrically by interleaving various definitions, theorems, and properties about vectors and matrices, linear systems and factorizations, eigenvalues and eigenvectors, singular values and singular vectors, linear transformations, orthogonality, determinants, etc.
5. Understand that many, seemingly disconnected, problems from various disciplines reduce to linear algebra problems.
6. Perform calculations with complex numbers.

Useful Matlab commands

	Matlab
Dot product	`dot`
Length of a vector	`norm`
Orthogonalize a matrix	`orth`
QR factorization	`QR`